NCERT · CLASS 7 · MATHS · PART II

Geometric Twins

C01

Congruence: same shape + same size

All four exercise sets · triangles · clear rules

Textbook exercise answers first · Illustrated explanations · Revision

Original learning notes based on the NCERT chapter. Diagrams are schematic and editable.

Part A · NCERT Figure it Out answers

Reading the exercise answers

The first chapter of *Ganita Prakash Part II* is printed as Chapter 1, so this is `Part_2/C01`. The descriptions below refer to the four *Figure it Out* sets (textbook pp. 3, 8–9, 13–14, 20–21). Use the original figures for visual comparison; the source PDF is not embedded.

Set 1 · Identical shapes and sizes (pp. 3–4)

Q1–Q2 · Compare the pictured pairs

Congruent figures fit **exactly** when traced and overlaid, including after turning or flipping. In Q1, the two figures are **not congruent**: the included angles differ even though the arm lengths look equal. In Q2 the two droplet outlines are congruent after rotation, and the two leaf pairs are congruent by reflection/rotation; the clouds and starbursts shown are of different sizes and are **not** congruent.

Q3 · Circle and rectangle

A circle is fixed by its **radius (or diameter)**. A rectangle is fixed by its **length and breadth**. To test congruence, compare these respective dimensions; orientation does not matter.

Q4 · Three-armed figures

Measure the three branch lengths from the junction and the **angles between branches**. Matching lengths alone do not guarantee the same figure. After measurements, rotate or flip one figure to test exact superposition. For the printed pairs, the **first pair appears congruent**, while the **second pair is not**; its angles or branch lengths differ. Check with a ruler/protractor to confirm on your printout.

Set 2 · Triangle correspondence (pp. 8–9)

Q1 · Other valid orders

From $\triangle HEN \cong \triangle BIG$, the five other equivalent statements are

$$\triangle HNE \cong \triangle BGI, \quad \triangle EHN \cong \triangle IBG,$$
$$\triangle ENH \cong \triangle IGB, \quad \triangle NHE \cong \triangle GBI,$$
$$\triangle NEH \cong \triangle GIB.$$

Corresponding vertices must stay in the same order on both sides.

Q2 · Two triangles with sides 3.5, 5 and 6 cm

The sides of $\triangle RED$ and $\triangle JAM$ match as follows: $RE = JA = 3.5$, $ED = AM = 5$, $RD = JM = 6$ cm. Hence

$$\boxed{\triangle RED \cong \triangle JAM \quad \text{(SSS)}}.$$

Q3 · Kite with $AB = AD$ and $CB = CD$

In triangles ABC and ADC: $AB = AD$, $BC = DC$, and AC is common. So $\triangle ABC \cong \triangle ADC$ by SSS. Therefore $\angle BAC = \angle CAD$ and $\angle BCA = \angle ACD$. Yes, AC bisects both indicated angles.

Q4 · $DF = DG$, $FE = GE$

Triangles DFE and DGE have $DF = DG$, $FE = GE$ and DE common. Thus $\triangle DFE \cong \triangle DGE$ by SSS. The triangles overlap after reflecting about DE.

Set 3 · Angles, sides and parallel lines (pp. 13–14)

Q1 · 7 cm, 5 cm and 47°

The 7-cm and 5-cm sides meet at the given 47° angle in both figures. Correspondence is $B \leftrightarrow Z$, $A \leftrightarrow X$, $C \leftrightarrow Y$.

$$\boxed{\triangle ABC \cong \triangle XZY \quad \text{(SAS)}}.$$

Q2 · $AB \parallel CD$ and $AB = CD$

The intersecting diagonals meet at O. Alternate angles from the parallel lines are equal, and the angles at O are vertically opposite. With $AB = CD$, triangles AOB and DOC are congruent by **ASA** (equivalently use their corresponding angles and the stated side). Therefore $AO = DO$ and $BO = CO$, along with equal corresponding angles.

Q3 · Equal pairs of angles at B and C

Given $\angle ABC = \angle DBC$ and $\angle ACB = \angle DCB$, third angles are equal because all triangle angles total 180°: $\angle BAC = \angle BDC$. Also BC is a common included side between the equal angles, so $\triangle ABC \cong \triangle DBC$ by ASA.

Q4 · Equal angles in crossed-triangle figure

Given $\angle ABD = \angle DCA$ and $\angle ACB = \angle DBC$. Add these angles on each side of the base:

$$\angle ABC = \angle ABD + \angle DBC = \angle DCA + \angle ACB = \angle DCB.$$

Thus $\angle ABC = \angle DCB$ and $\angle ACB = \angle DBC$, while BC is common. So

$$\boxed{\triangle ABC \cong \triangle DCB \quad \text{(ASA)}}.$$

Consequently $AB = DC$, $AC = DB$, and $\angle BAC = \angle CDB$.

Set 4 · Final exercise (pp. 20–21)

Q1 · $\triangle AIR \cong \triangle FLY$

Vertices: $A \leftrightarrow F$, $I \leftrightarrow L$, $R \leftrightarrow Y$.
Sides: $AI = FL$, $IR = LY$, $AR = FY$. Angles: $\angle A = \angle F$, $\angle I = \angle L$, $\angle R = \angle Y$.

Q2 · Which measurements guarantee congruence?

(a) Yes: SSS, $\triangle ABC \cong \triangle DEF$.
(b) Yes: SAS, $\triangle ABC \cong \triangle EFD$.
(c) Yes: RHS, $\triangle ABC \cong \triangle FDE$.
(d) Yes: AAS, $\triangle ABC \cong \triangle DEF$.
(e) Not guaranteed: SSA (non-included angle) may give two distinct triangles.

Q3 · Prove $AB \parallel CD$

As $AO = OD$, $BO = OC$, and $\angle AOB = \angle DOC$ (vertically opposite), $\triangle AOB \cong \triangle DOC$ by SAS. Hence $\angle BAO = \angle CDO$. These are alternate angles made by transversal AD, so $\boxed{AB \parallel CD}$.

Q4 · Square cut along diagonal

For square $ABCD$, $AB = AD$, $BC = DC$, and AC is common; hence $\triangle ABC \cong \triangle ADC$ (SSS). Also $\triangle ABC \cong \triangle CDA$ (the two 45-degree corners interchange). Two identical **isosceles right triangles** allow two valid vertex correspondences; two equilateral triangles allow **six** because all three vertices are interchangeable.

Q5 · Circle with central angle 120°

The two radii $AB = AC$, so triangle ABC is isosceles. $\angle B = \angle C = (180° - 120°)/2 = \boxed{30°}$.

Q6 · Large marked-angle construction

The textbook diagram on p. 21 has four unlabelled interior points. For clarity, call them S, T, Q, P as follows: S is the junction below R; T the junction below U; Q the central junction below V; and P the lower junction directly right of T. In the table, angle triples are the three measures for each triangle (not necessarily in vertex order).

Triangle	Angles (°)	**Triangle**	Angles (°)
CRU	$90, 45, 45$	TPK	$102, 30, 48$
RUS	$45, 33, 102$	KPL	$30, 60, 90$
UST	$34, 73, 73$	PLB	$90, 60, 30$
UTA	$68, 56, 56$	PBF	$60, 60, 60$
ATK	$34, 44, 102$	PFQ	$94, 22, 64$
RSQ	$34, 102, 44$	QFD	$98, 56, 26$
STQ	$83, 51, 46$	QDV	$112, 34, 34$
TQP	$34, 90, 56$	QVR	$68, 56, 56$

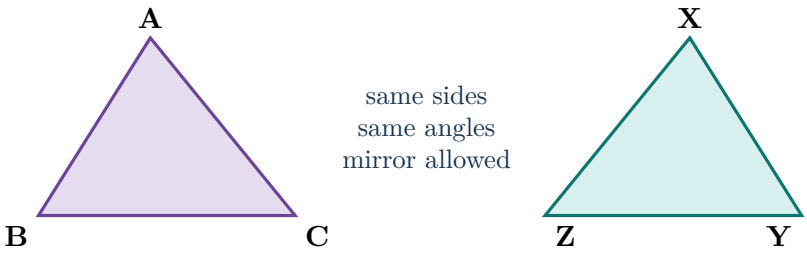
Reasoning: First locate isosceles triangles using equal-side tick marks; their base angles are equal. Use angles summing to 180° in each triangle, supplementary angles on straight lines, and 360° around a point. As a check, the angle sectors around Q sum to $44 + 46 + 90 + 64 + 26 + 34 + 56 = 360°$. Use the original figure while assigning the temporary labels; it has no printed P, Q, S, T labels.

Key textbook activities · reasoning

- **SSS:** a triangle with sides $4, 6, 8$ cm is fixed up to reflection. A mirror copy is still congruent.
- **SAS:** sides $6, 5$ cm with the *included* 30° angle fix a triangle.
- **SSA:** sides $6, 4$ cm and a *non-included* 30° angle can produce two non-congruent triangles.
- **ASA:** included side 5 cm with adjacent 50° and 30° angles fixes a triangle.
- **AAS:** with angles $35°, 75°$, third angle is 70°; a corresponding side fixes the triangle.
- **RHS:** right angle, hypotenuse 5 cm and one leg 4 cm fix the triangle.

Part B · See the geometry

A triangle can rotate or reflect yet remain congruent. A diagram illustrates the matching order:



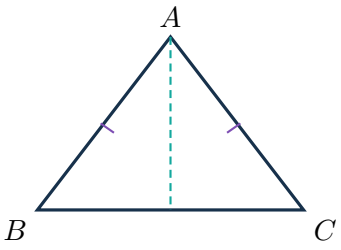
Five conditions that work:

Code	Required matching information
SSS	Three pairs of equal sides
SAS	Two pairs of equal sides and their included angle
ASA	Two pairs of equal angles and the included side
AAS	Two pairs of equal angles and a corresponding non-included side
RHS	Both triangles right-angled; equal hypotenuse and one equal leg

Why SSA cannot be used as a general test

Two sides and an angle *not between them* may allow the third vertex to be at two different intersections of the same circle and ray. Two possible triangles can therefore satisfy the information. Do not confuse SSA with SAS.

Isosceles and equilateral facts:



$AB = AC$
$\angle B = \angle C$
if $\angle A = 80°$:
$\angle B = \angle C = 50°$

Common mistakes

AAA shows equal angles (same shape), not necessarily equal sizes. SSA does not generally guarantee congruence. Put letters in matching vertex order when writing $\triangle ABC \cong \triangle XYZ$. Equal sides face equal angles, not adjacent ones.

Memory check

1. Two triangles have side lengths 3, 4, 5 cm each. Congruent? *Yes, SSS.*
2. Two triangles have angles $40°, 60°, 80°$. Congruent? *Not necessarily, AAA.*
3. In an isosceles triangle the vertex angle is $100°$. Base angles? $40°$ *each.*
4. In a right triangle, hypotenuse and one leg agree. Which rule? *RHS.*

For a geometry proof, always write **Given** → **Equal parts** → **Rule** → **Conclusion**.